Gamma-ray burst classification

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Introduction to Bayesian probability theory

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The Gamma Ray Bursts (GRBs)

They are transient events of high energy photons and extragalactic origin, discovered in 1973

They are usually composed of a first peak (prompt emission) and then an afterglow

We will use two proprieties to classify them:

- The T_{90}: the time needed to pass from 5% to 95% of the total flux of the prompt emission

- The HR: the ratio between fluxes in different energy bands

Tipical light curve of a GRB?
Or artistic representation?

Classification on $\log(T_{90})$: data

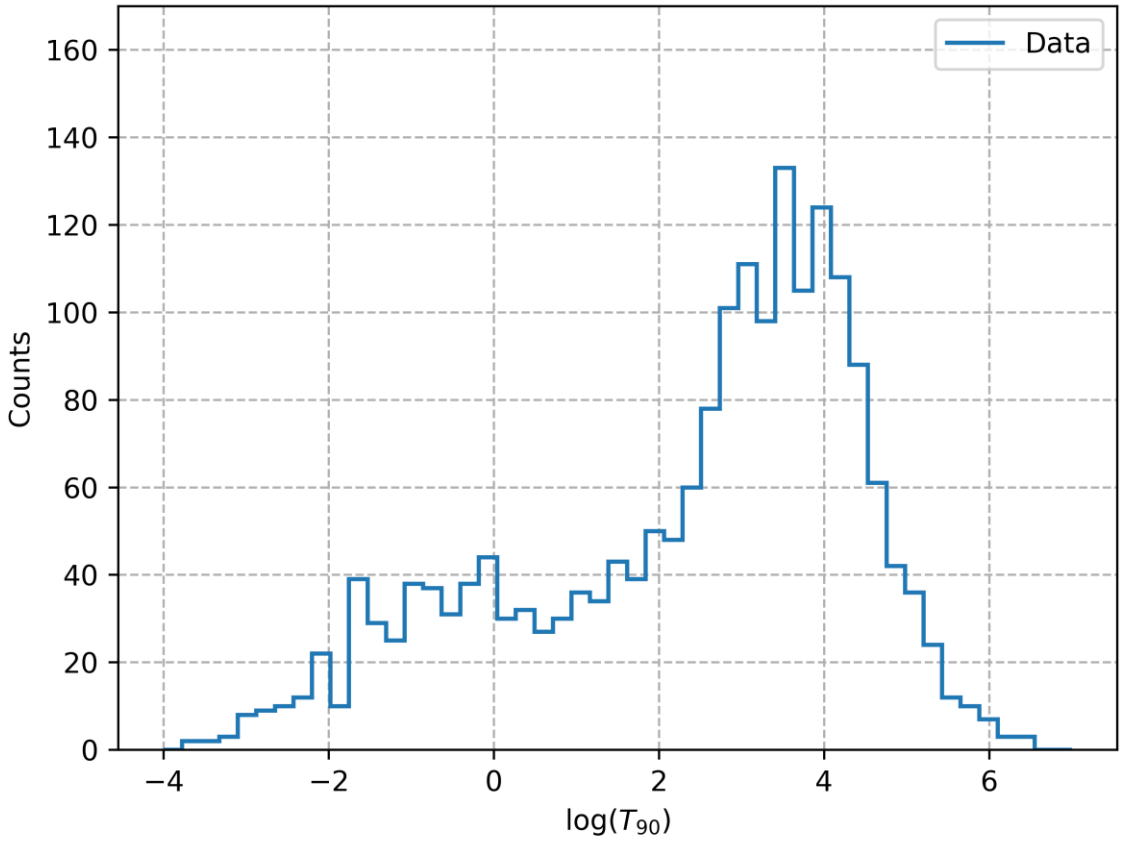
Dataset: 1932 burst seen by BATSE with the recorded $\log(T_{90})$, associated error and $\log(HR)$

Histogram on $\log(T_{90})$

Smallest value: ~ -3.7
Biggest value: ~ 6.5

50 bins from -4 to 7

Mean error is: 0.174
Median error is: 0.0919
A bin is large 0.224



Classification on $\log(T_{90})$: model

The model for the classification is the weighted sum of two log normal distribution

$$p(\log(T_{90})) = w\, N(\log(T_{90})\,|\mu_1\sigma_1) + (1-w)\, N(\log(T_{90})\,|\mu_2\sigma_2)$$

So we have 5 parameters to estimate, we will call them $\theta \equiv [w,\, \mu_1,\, \sigma_1,\, \mu_2,\, \sigma_2]$

Classification on $\log(T_{90})$: Bayes theorem

To estimate the parameters we will use Bayes theorem, define the propositions

- θ = «The set of parameters is θ»
- D = «The dataset we are analysing»
- H = «Model of the two weighted sum log normal»
- I = «All the background information on GRBs»

$$P(\theta|HDI) = P(\theta|HI)\frac{P(D|\theta HI)}{P(D|HI)}$$

Or in the log form:

$$\log P\,(\theta|HDI) = \log P\,(\theta|HI) + \log P\,(D|\theta HI) - \log P\,(D|HI)$$

Log prior — Log likelihood — Normalization costant

Classification on $\log(T_{90})$: Prior

$P(\theta|HI)$ Probability for a set of parameters given the model and background. Outside of the bounds is zero, inside?

For the weight and the means we chose a uniform prior

$$0 < w < 1$$

$$-4 < \mu_i < 7$$

$$Prior \propto 1$$

For the sigmas we chose a Jeffrey's prior because this are scale parameters of the problem

$$0.01 < \sigma_i < 3$$

$$Prior \propto 1/\sigma$$

Classification on $\log(T_{90})$: Likelihood

$P(D|\theta HI)$ Probability of seeing the dataset given a set of parameters, the model and background

Having the model, the parameters and the bins we can compute the predicted count in a bin by our model

$$\lambda_b = N \cdot p(\log(T_{90}))_b \cdot db$$

The probability of having the count c_b in a bin given the predicted count λ_b is the Poisson's distribution

$$P(c_b|\lambda_b) = \frac{1}{c_b!}\lambda_b^{c_b} e^{-\lambda_b}$$

The likelihood is the sum over all the bins!

$$P(D|\theta HI) = \sum_b P(c_b|\theta HI)$$

Sample the posterior: Metropolis-Hasting

Given a set of parameters we can now compute the posterior. How we find the set that maximize it?

5D grid search: not smart

Metropolis-Hasting algorithm!

$q(\theta)$ is the proposal distribution

We chose a 5D multivariate gaussian but how we chose his covariance matrix?

- «Blind» run: we use a diagonal covariance matrix with elements equals to 5% of the bound
- «Not Blind» run: we use a covariance matrix estimated after a previous run

Base idea

- Sample a θ_t from $q(\theta_0)$

Sample the posterior: Metropolis-Hasting

Given a set of parameters we can now compute the posterior. How we find the set that maximize it?

5D grid search: not smart

Metropolis-Hasting algorithm!

Acceptance rule:

$$\frac{p(\theta_t)}{p(\theta_0)} \cdot \frac{q(\theta_t|\theta_0)}{q(\theta_0|\theta_t)} \geq a$$

Posterior values

Proposal probability ratio
ours is symmetric so it's one

Random number
from [0,1]

Base idea

- Sample a θ_t from $q(\theta_0)$
- Compute the posterior on θ_t
- Check acceptance rule

Sample the posterior: Metropolis-Hasting

Given a set of parameters we can now compute the posterior. How we find the set that maximize it?

5D grid search: not smart

Metropolis-Hasting algorithm!

We make a first «blind» run of the algorithm with 10k steps to estimate the covariance

We then make the final run with 100k steps

Base idea

- Sample a θ_t from $q(\theta_0)$
- Compute the posterior on θ_t
- Check acceptance rule

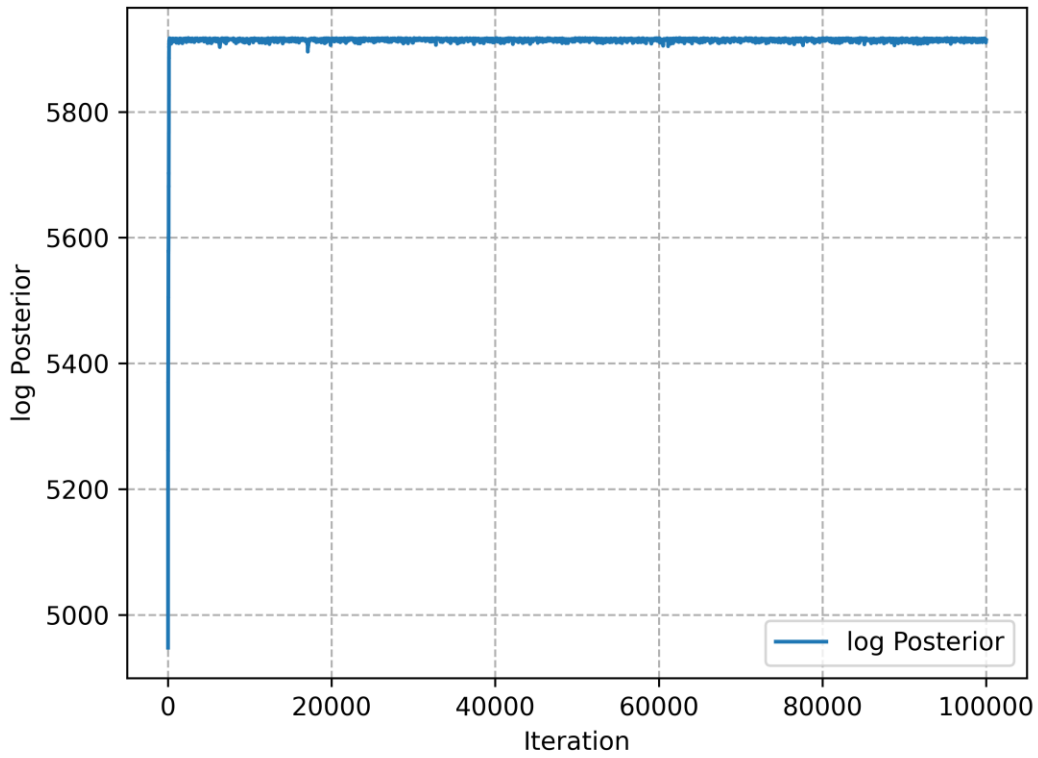
If true move to $\theta_0 = \theta_t$

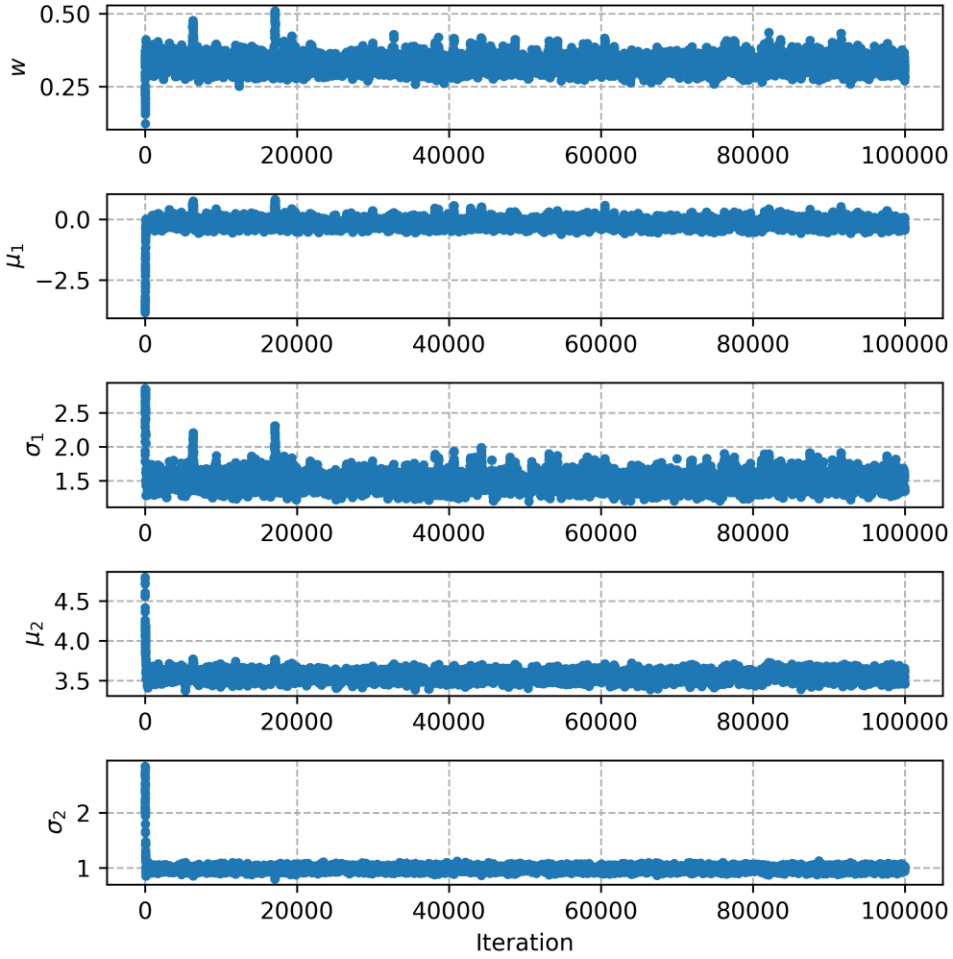
If false stay at θ_0

In this way we will have the θ with higher posterior more times so more probable!

Classification on $\log(T_{90})$: traces

We see that some steps are needed to find the peak → Burn-in: discard n points to have samples of the peak

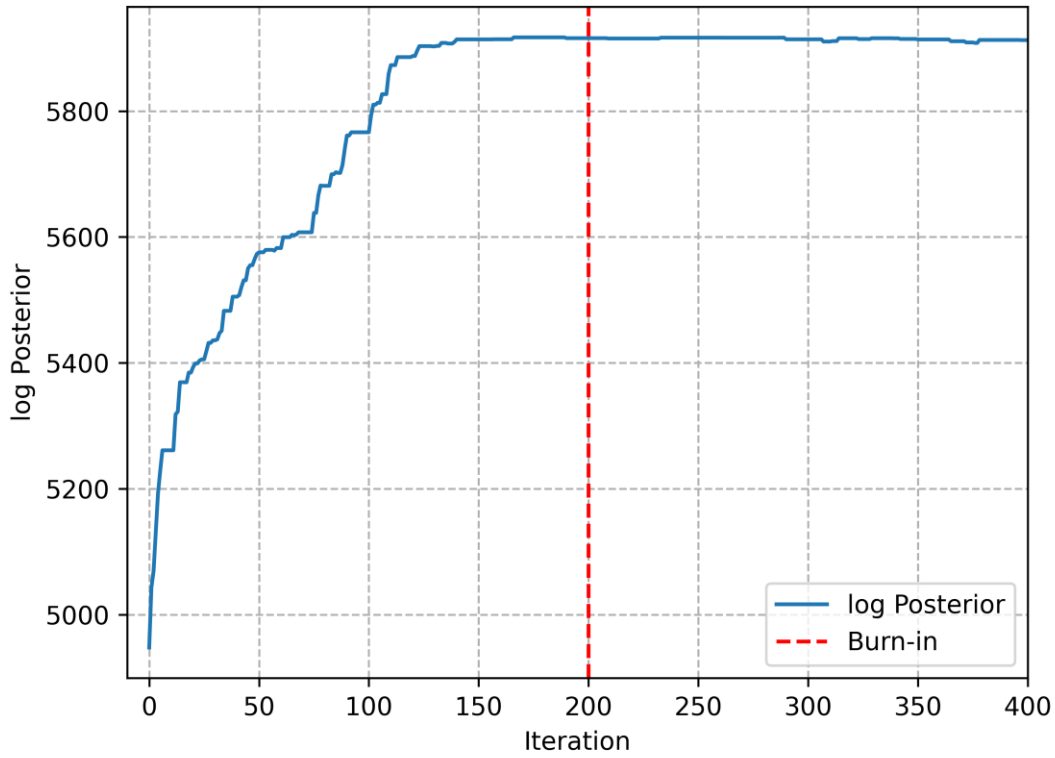


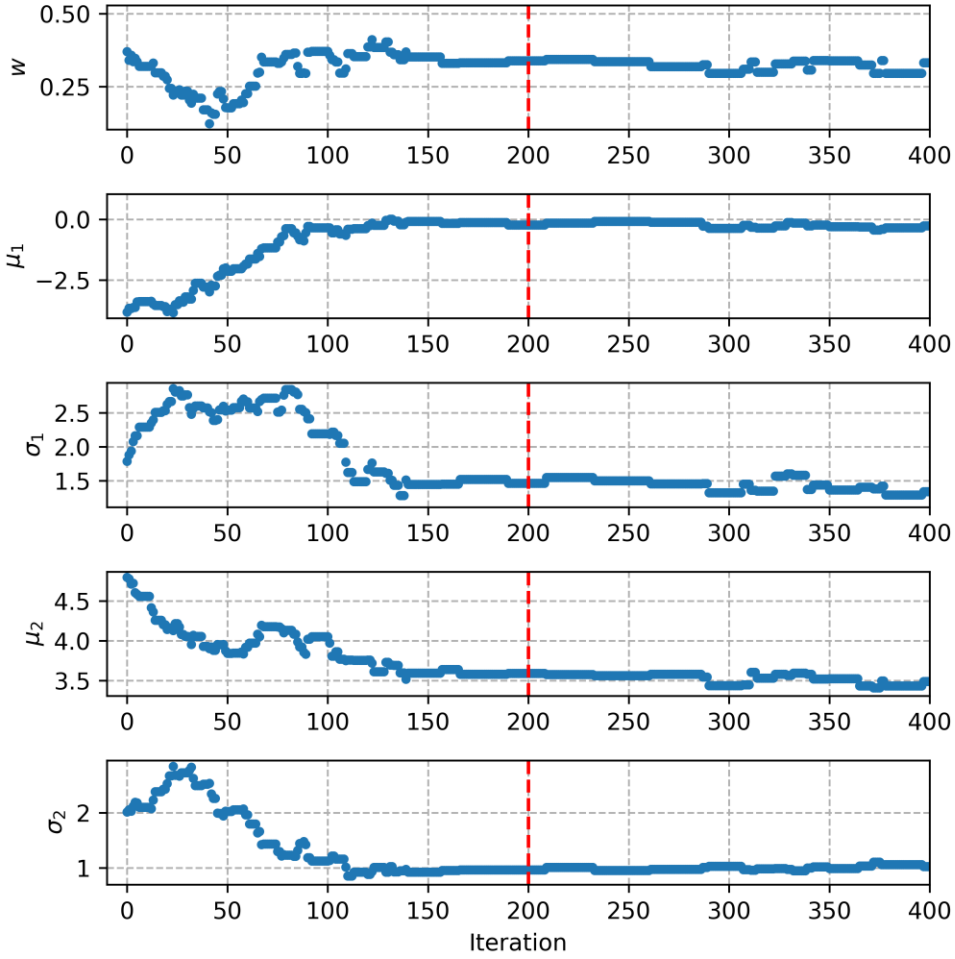


Classification on $\log(T_{90})$: traces

We see that some steps are needed to find the peak → Burn-in: discard n points to have samples of the peak

Burn-in = 200

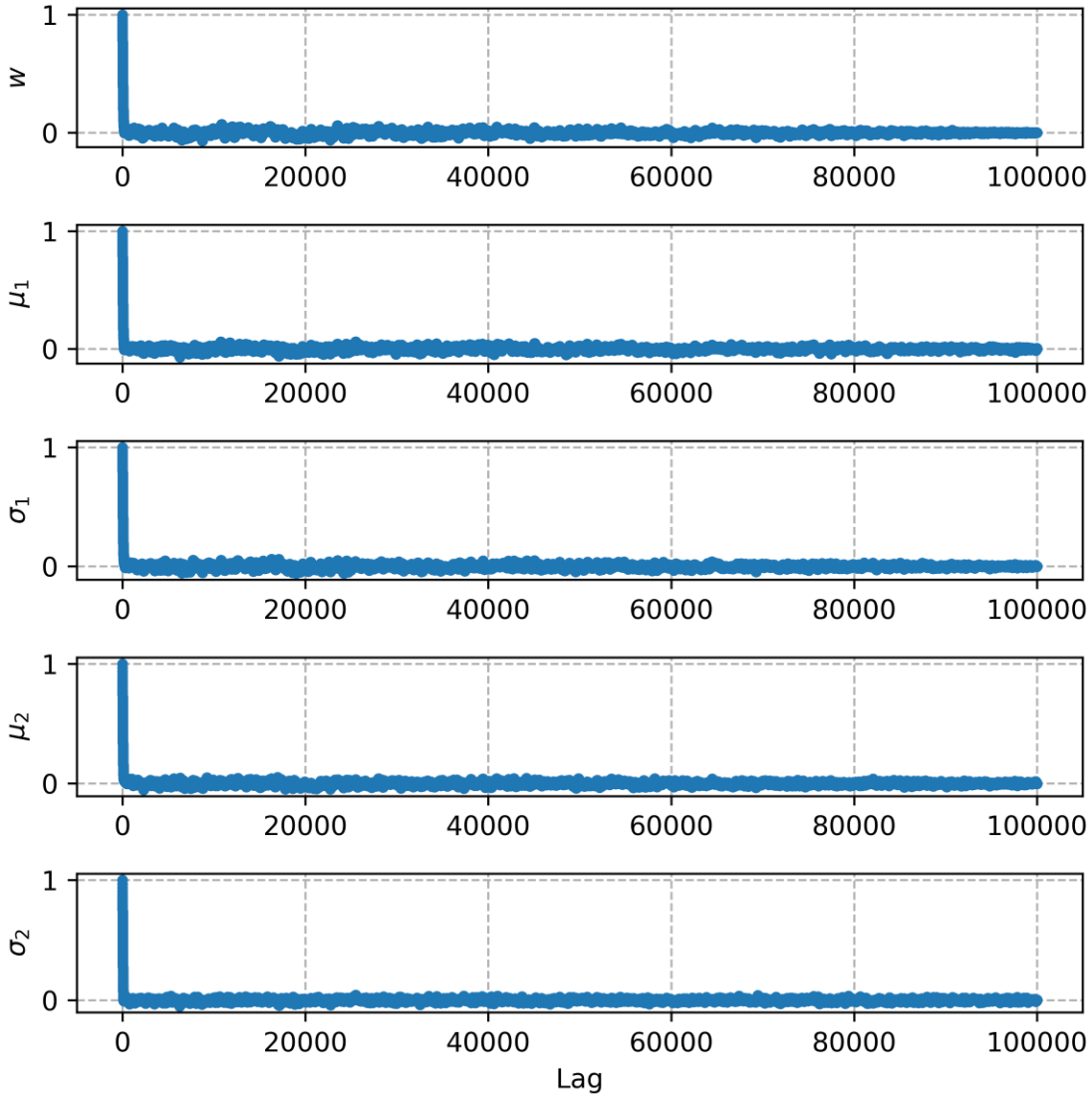




Classification on $\log(T_{90})$: autocorrelation

We look at the autocorrelation for the parameters and we see that for small lag the samples are correlated

Thinning: take a point every n to have not correlated samples

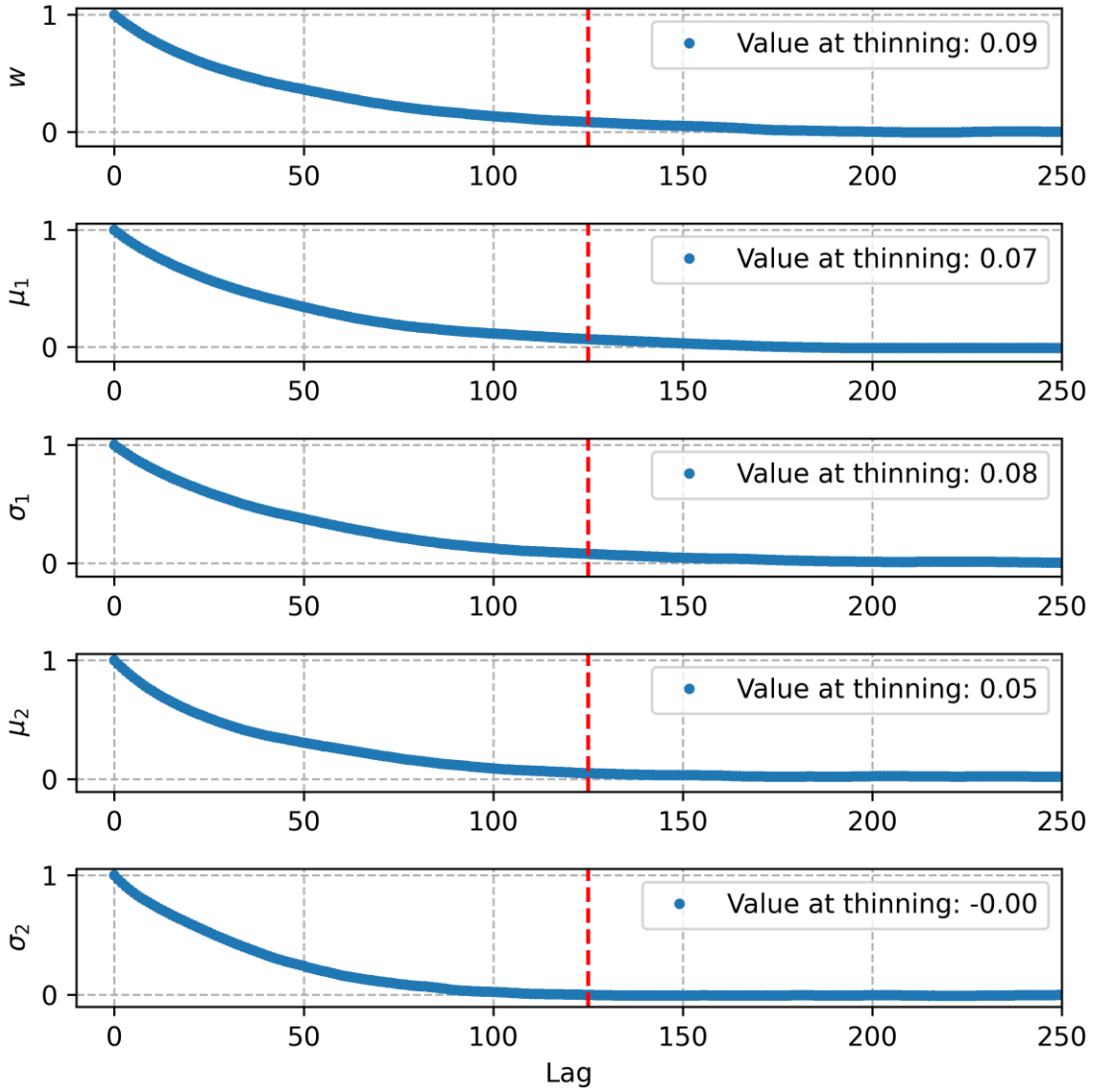


Classification on $\log(T_{90})$: autocorrelation

We look at the autocorrelation for the parameters and we see that for small lag the samples are correlated

Thinning: take a point every n to have not correlated samples

Thinning = 125



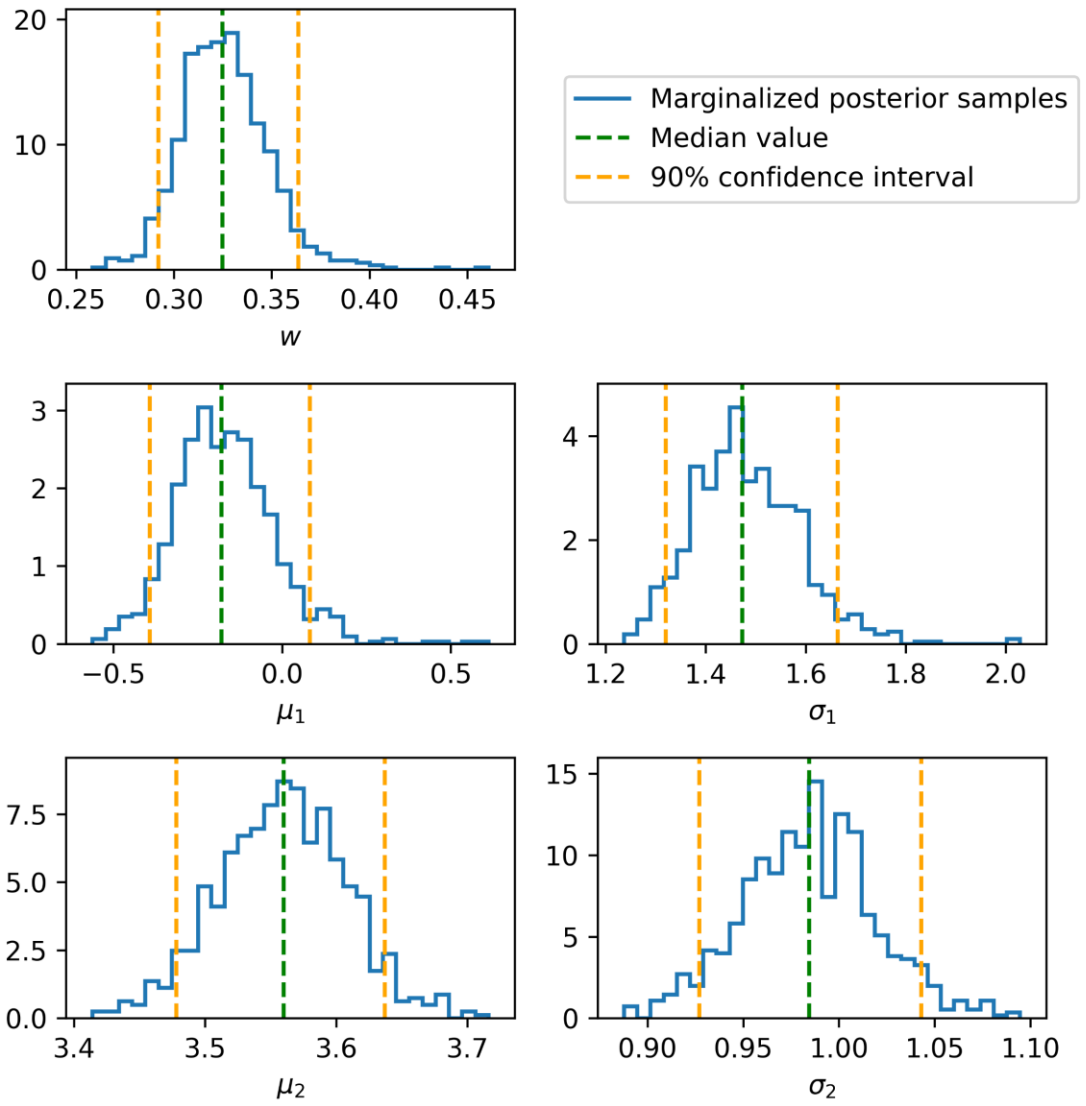
Classification on $\log(T_{90})$: marginals

We make the histograms for each parameter
We find the median value and the interval corresponding to 90% level of confidence

w
$0.325^{+0.039}_{-0.033}$

μ_1	σ_1
$-0.18^{+0.26}_{-0.21}$	$1.47^{+0.19}_{-0.15}$

μ_2	σ_2
$3.560^{+0.077}_{-0.082}$	$0.984^{+0.058}_{-0.057}$



w	μ_1	σ_1	μ_2	σ_2
$0.326^{+0.024}_{-0.020}$	$-0.18^{+0.15}_{-0.14}$	$1.48^{+0.12}_{-0.10}$	$3.56^{+0.048}_{-0.050}$	$0.982^{+0.035}_{-0.033}$

Classification on $\log(T_{90})$: final model

Finally we look at the final model, computed as the median of the models computed on each sample (and not the model on the median values) and we also plot the interval corrisponding to 68% level of confidence. We also show the two separete components of the model

Final model img

Classification on $\log(T_{90})$ with errors

Consider

Conclusions

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